

Users' Perception Analysis of Cyberattack Consequences Using NLP

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CS 273 Project 2.2

I. INTRODUCTION

Project Overview: The main goal of this project was to compare human judgments of cyberattack consequences with similarity values produced by NLP methods. The human data came from a card-sorting process, while the NLP part used the text of the 50 consequence descriptions. We wanted to see whether consequences that people grouped in similar ways would also look similar when they were compared using BOW, TF-IDF, or WordNet.

The project had three main experiments. First, we created the human-based similarity matrix using Spearman correlation. Second, we created 12 BOW matrices and 12 TF-IDF matrices using the required preprocessing configurations and cosine similarity. Third, we created four WordNet matrices using the required GLAO-style procedure. We then compared all 28 NLP matrices with the human matrix using Pearson correlation.

II. DATA DESCRIPTION AND LIBRARIES

The data came from the provided `all_data.xlsx` file. The worksheet used in the analysis contains 226 rows and 53 columns. The first two columns identify the participant and group, and the remaining 50 columns are the cyberattack consequence descriptions. Each consequence column is also a 226-value binary vector created from the card-sorting data. A value of 1 shows that the consequence was included in the group represented by that row, while 0 shows that it was not. This allowed us to use the column vectors for the human experiment and the column names as the text documents for the NLP experiments.

The notebook was written in Python. The main libraries used were NumPy and pandas for data handling, SciPy for Spearman and Pearson correlation, scikit-learn for CountVec-torizer, TfidfVectorizer, and cosine similarity, NLTK for tokenization, POS tagging, stopwords, stemming, lemmatization, and WordNet, and Matplotlib for the heatmaps. We also used utilities from `itertools`, `functools`, and `IPython` to organize repeated configurations and display the results.

III. EXPERIMENT 1: HUMAN-BASED SIMILARITY MATRIX

For the first experiment, each cyberattack consequence was represented by its provided 226-value card-sorting vector. We

compared every pair of consequence vectors using Spearman's rank correlation. The calculation can be written as

$$\rho_s(x, y) = \text{corr}(R(x), R(y)), \quad (1)$$

where $R(x)$ and $R(y)$ are the ranked versions of the two vectors. We used SciPy's implementation because the binary vectors contain many tied values.

The result was a symmetric 50×50 matrix with 1 on the diagonal. As shown in Fig. 1(a), the human matrix contains both positive and negative relationships. It also contains several visible blocks where groups of consequences were treated similarly by participants.

IV. EXPERIMENT 2: BOW AND TF-IDF

A. Preprocessing Configurations

For BOW and TF-IDF, each consequence description was converted to lowercase, tokenized, and limited to alphabetic tokens. We then tested all combinations of three preprocessing choices: stopwords kept or removed, stemming off or on, and unigrams, bigrams, or trigrams. This produced $2 \times 2 \times 3 = 12$ configurations for BOW and the same 12 configurations for TF-IDF.

BOW represents each consequence using counts of the selected words or n -grams. TF-IDF uses the same general features but gives less weight to features that appear in many documents. After vectorization, every pair of consequence vectors was compared using cosine similarity,

$$\cos(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\|_2 \|\mathbf{y}\|_2}. \quad (2)$$

Each configuration therefore produced its own 50×50 similarity matrix.

V. EXPERIMENT 3: WORDNET SIMILARITY

The WordNet experiment followed the required seven-step process. The purpose was to compare the sentences semantically instead of relying only on exact word overlap. The steps used in the notebook were:

- 1) Tokenize each sentence and assign Penn Treebank POS tags with `word_tokenize()` and `pos_tag()`.
- 2) Convert the Penn tags to WordNet tags `n`, `v`, `a`, or `r`. Tag `s` was treated as `a`, and unsupported tags were removed.
- 3) Retrieve the first WordNet synset for each supported token using `wn.synsets()`.
- 4) Compare synset pairs using WordNet `path_similarity()`.

- 5) For each source synset, keep its best target match, sum the valid scores, and divide by the maximum number of synsets in the two sentences.
- 6) Compute the score in both directions and average the two values so the result is symmetric.
- 7) Repeat the process for every pair of the 50 consequence descriptions to build the similarity matrix.

If A and B are the two synset lists, the directional score is

$$S(A, B) = \frac{\sum_{a \in A} \max_{b \in B} \text{path}(a, b)}{\max(|A|, |B|)}, \quad (3)$$

and the final symmetric score is

$$S_{\text{sym}}(A, B) = \frac{S(A, B) + S(B, A)}{2}. \quad (4)$$

We tested four WordNet configurations: stopwords kept or removed, crossed with lemmatization off or on. This produced four WordNet similarity matrices.

VI. MATRIX COMPARISON

To compare each NLP matrix with the human matrix, we extracted the upper triangle and excluded the diagonal. With 50 consequences, this gives 1,225 unique consequence pairs. We used the same pair positions for every matrix and calculated Pearson correlation,

$$r = \frac{\sum_i (h_i - \bar{h})(n_i - \bar{n})}{\sqrt{\sum_i (h_i - \bar{h})^2} \sqrt{\sum_i (n_i - \bar{n})^2}}, \quad (5)$$

where h contains the human similarity values and n contains the NLP similarity values. A larger positive value means that the NLP matrix follows the human pairwise pattern more closely.

VII. RESULTS

Figure 1 shows the human matrix and the best matrix from each NLP technique. The consequence order is the same in all four panels, which makes the main blocks easier to compare visually.

A. BOW Results

Table I shows all 12 BOW configurations. The highest correlation was **0.599**, using stopword removal, no stemming, and unigrams. Stemming changed the unigram results only slightly. Stopword removal improved the unigram result, but it made the bigram and trigram results much lower. This suggests that the individual content words were more useful than longer word sequences for this dataset.

B. TF-IDF Results

Table II shows all 12 TF-IDF configurations. The highest TF-IDF correlation was **0.555**, also using stopword removal, no stemming, and unigrams. The general pattern was similar to BOW. Unigrams performed better after stopword removal, while the bigram and trigram versions became weaker. BOW was higher than TF-IDF in the best configuration, which suggests that some of the repeated vocabulary across the consequence descriptions was still useful for matching the human groupings.

TABLE I
BOW CORRELATION WITH THE HUMAN MATRIX

Stopwords	Stemming	n -gram	Pearson r
Kept	No	Unigram	0.522
Kept	No	Bigram	0.529
Kept	No	Trigram	0.415
Kept	Yes	Unigram	0.521
Kept	Yes	Bigram	0.526
Kept	Yes	Trigram	0.413
Removed	No	Unigram	0.599
Removed	No	Bigram	0.382
Removed	No	Trigram	0.172
Removed	Yes	Unigram	0.595
Removed	Yes	Bigram	0.378
Removed	Yes	Trigram	0.172

TABLE II
TF-IDF CORRELATION WITH THE HUMAN MATRIX

Stopwords	Stemming	n -gram	Pearson r
Kept	No	Unigram	0.497
Kept	No	Bigram	0.487
Kept	No	Trigram	0.396
Kept	Yes	Unigram	0.489
Kept	Yes	Bigram	0.481
Kept	Yes	Trigram	0.394
Removed	No	Unigram	0.555
Removed	No	Bigram	0.386
Removed	No	Trigram	0.173
Removed	Yes	Unigram	0.538
Removed	Yes	Bigram	0.380
Removed	Yes	Trigram	0.173

C. WordNet Results

Table III shows the four WordNet configurations. Removing stopwords increased the correlation from about 0.525 to **0.558**. Lemmatization did not change the final correlation, so both stopword-removed configurations tied for the highest WordNet result.

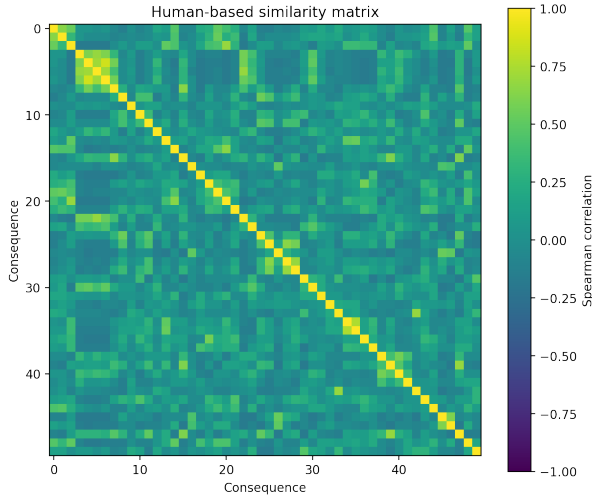
TABLE III
WORDNET CORRELATION WITH THE HUMAN MATRIX

Stopwords	Lemmatization	Pearson r
Kept	No	0.525
Kept	Yes	0.525
Removed	No	0.558
Removed	Yes	0.558

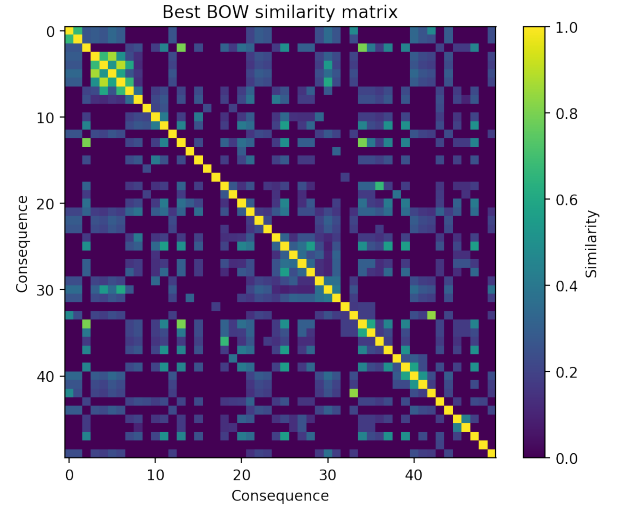
D. Overall Interpretation

The strongest result across all 28 NLP matrices was BOW at 0.599. WordNet reached 0.558 and TF-IDF reached 0.555. The differences are not extremely large, but the results show that simple word overlap worked very well for these short consequence descriptions. Figure 1 also shows that BOW and TF-IDF are relatively sparse, while WordNet is denser because related words can receive a positive semantic score even when the exact words are different.

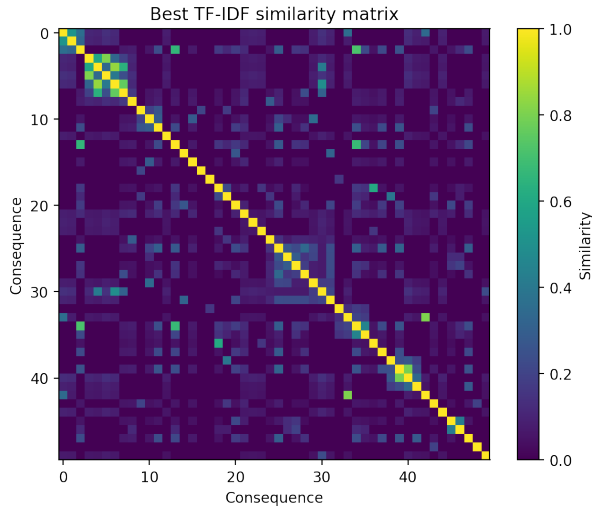
The preprocessing results were also useful. Stopword removal helped the best unigram configurations for all three



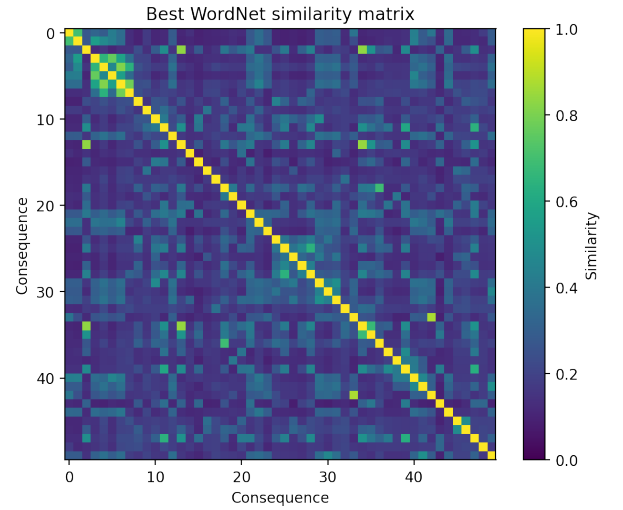
(a) Human-based similarity matrix



(b) Best BOW similarity matrix



(c) Best TF-IDF similarity matrix



(d) Best WordNet similarity matrix

Fig. 1. Human-based similarity matrix and the best BOW, TF-IDF, and WordNet matrices.

techniques, but it caused large decreases for BOW and TF-IDF bigrams and trigrams. This shows why testing all required configurations was important instead of assuming that one preprocessing choice would always improve the result. One limitation is that the human matrix can contain negative Spearman values, while the NLP similarities used here are nonnegative. Also, the WordNet procedure uses the first synset and does not perform word-sense disambiguation, so the results should be interpreted for this dataset rather than as a general ranking of NLP methods.

VIII. CONCLUSION

This project compared human card-sorting similarity with BOW, TF-IDF, and WordNet similarity for 50 cyberattack consequence descriptions. We created one human matrix, 24 BOW/TF-IDF matrices, and four WordNet matrices, then compared all 28 NLP matrices with the human matrix using Pearson correlation on the upper triangle.

The highest BOW result was 0.599 with stopwords removed, no stemming, and unigrams. TF-IDF reached 0.555 with the same setup, and WordNet reached 0.558 after stopword removal. These results show that simple unigram word overlap followed the human grouping pattern especially well for this dataset.

IX. PROJECT LINK

<https://www.overleaf.com/read/wnzdsddhtyjg#6b8116>